A Simple Example of Brenier's Theorem

Consider two probability measures on $\mathbb{R}$:

$$\mu = \mathrm{Unif}[0,1], \qquad \nu = \mathrm{Unif}[1,2].$$

The optimal transport map for the quadratic cost

$$c(x,y) = |x-y|^2$$

is

$$T(x) = x + 1.$$

Indeed, T pushes μ forward to ν:

$$T_{\#}\mu = \nu,$$

because translating a uniform random variable on $[0,1]$ by 1 gives a uniform random variable on $[1,2]$.

Brenier Form

Brenier's theorem says that, under its hypotheses, the optimal map has the form

$$T = \nabla\varphi$$

for a convex function φ.

In one dimension, the gradient is just the derivative. Here

$$T(x) = x + 1$$

is the derivative of

$$\varphi(x) = \frac{1}{2}x^2 + x.$$

Since

$$\varphi''(x) = 1 > 0,$$

the function φ is convex. Therefore

$$T(x) = \varphi'(x) = x + 1$$

is exactly a Brenier map.

Interpretation

This example says that the optimal way to move the uniform distribution on $[0,1]$ to the uniform distribution on $[1,2]$ is to translate every particle one unit to the right.